JM0211 Process Mining

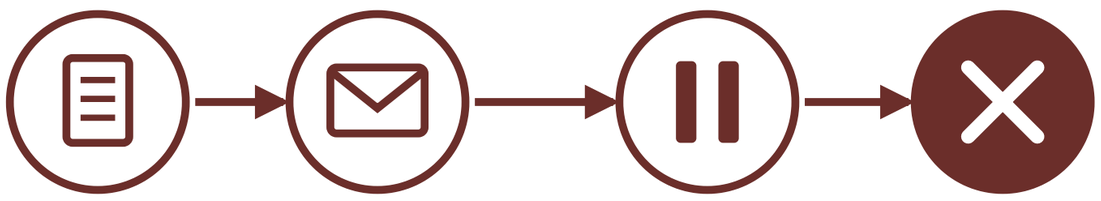
Conformance Checking for Process Improvement

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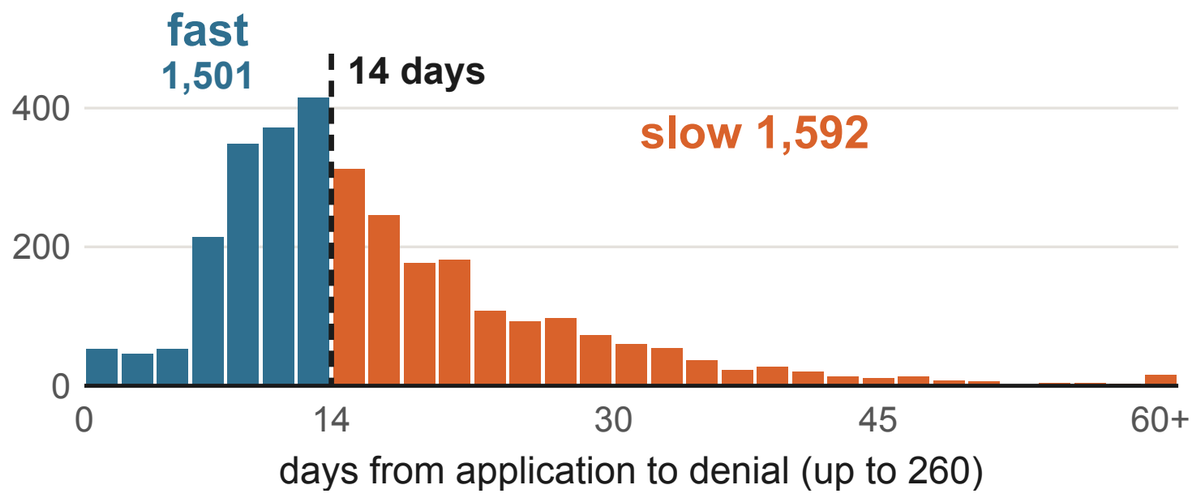
Jheronimus Academy of Data Science

Motivation and Scientific Background

BPI 2017 · Dutch bank · **3,093 denied loans**



Which deviations from the process go with slow denials?



Explanation of Conceptual Approach

Input — event log + Petri net

1 KPI — days per case

2 Align — count deviations

3 Tree — slow vs fast

Output — rules + cases

Case	Call paused	**Call paused**	>>	Denied
Model	Call paused	>>	**Call resumed**	Denied
	✓	**log move**	**model move**	✓

log move extra step in the case

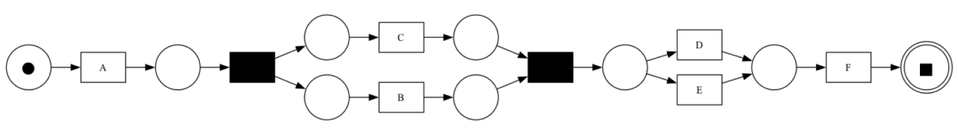
model move step the case skipped

One case → 108 deviation counts → tree

Evaluation Design and Results

① Is it correct?

Toy logs, solved by hand first



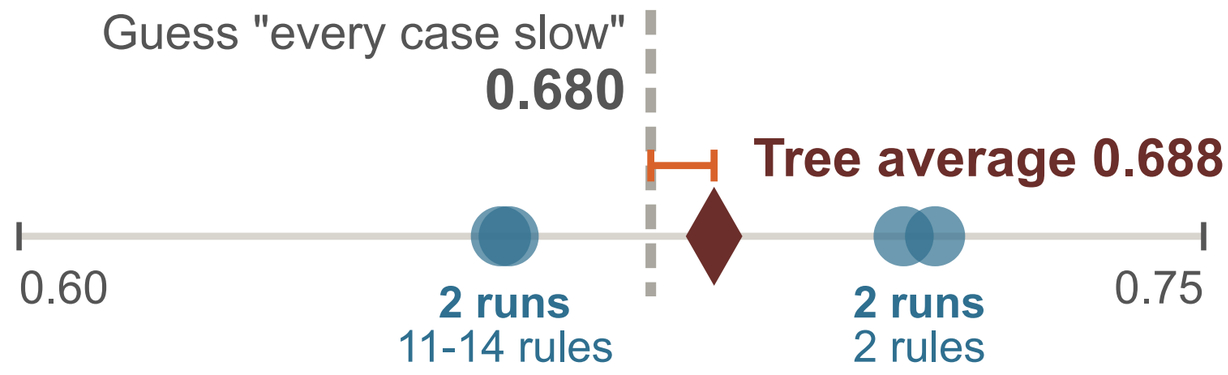
✓ **Yes, exact match**

11 / 11 alignments

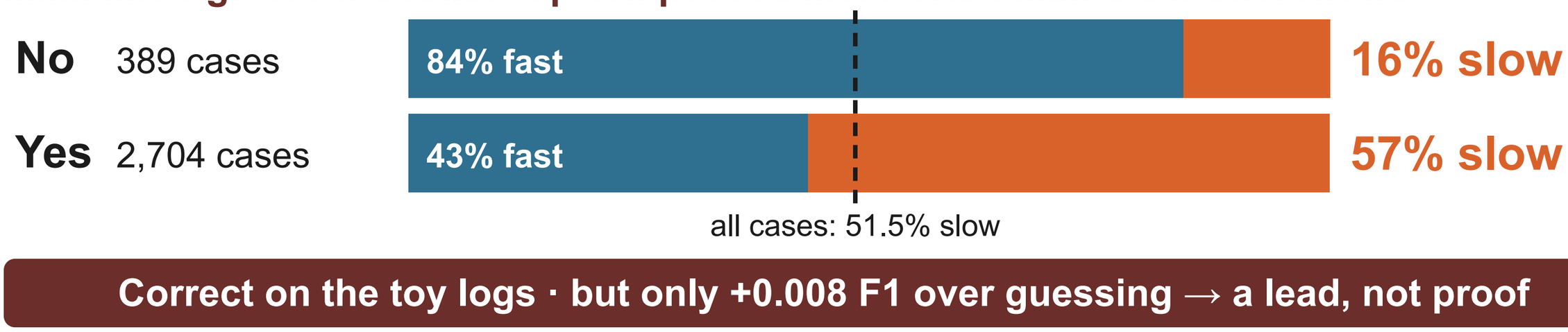
2 / 2 planted rules

② Is it useful? Barely better than guessing

4 runs: trained on a random 75%, tested on the other 25%



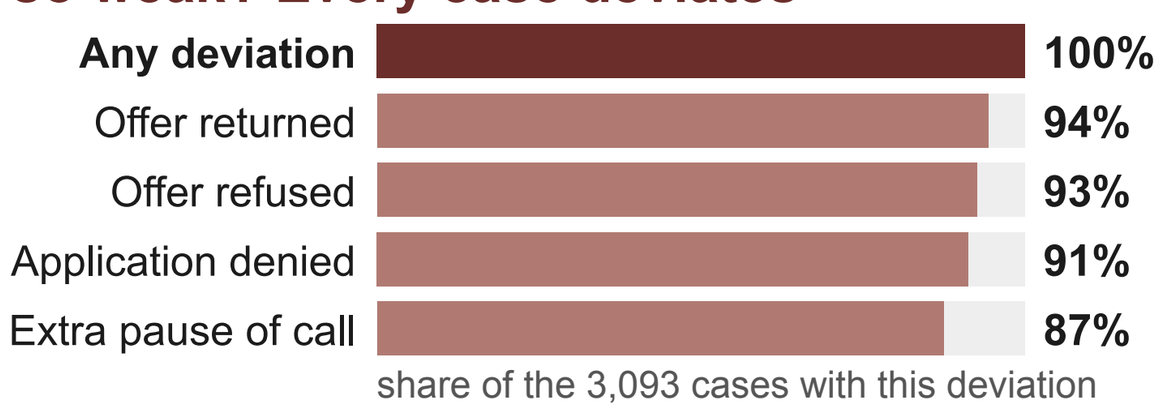
Main finding: was the follow-up call paused more often than the model allows?



Correct on the toy logs · but only +0.008 F1 over guessing → a lead, not proof

Limitations and Future Work

Why so weak? Every case deviates



Model too strict: deviations are normal, so they hardly tell slow from fast.

- **Net that allows repeated pauses**
- **Add waiting-time features**
- **Predict days, not a cutoff**
- **Also check data (loan amount)**

References: Dees, de Leoni, Mannhardt (2017). van der Aalst (2016) Process Mining. Berti et al. (2019) PM4Py. van Dongen (2017).